

EXPERIMENT 1

Aim

To calculate the average marks of each subject and identify the subject with the highest average.

Algorithm

1. Import the NumPy library.
2. Create a 2D array containing student marks.
3. Calculate subject averages using `np.mean()`.
4. Find the highest average using `np.argmax()`.
5. Display the results.

Code

```
import numpy as np

marks = np.array([
    [85, 90, 78, 88],
    [75, 80, 85, 70],
    [90, 95, 88, 92],
    [80, 85, 90, 82] ])

subjects = ["Math", "Science", "English", "History"]

averages = np.mean(marks, axis=0)

for i in range(4):
    print(subjects[i], "Average:", averages[i])

index = np.argmax(averages)

print("Highest Average Subject:", subjects[index])
print("Highest Average:", averages[index])
```

Input

Student marks in four subjects: Math, Science, English, and History.

OUTPUT:

```
File Edit Shell Debug Options Window Help
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 1.py =====
AVERAGE MARKS OF EACH SUBJECT
Math : 82.5
Science : 87.5
English : 85.25
History : 83.0
Subject with highest average: Science
Highest average: 87.5
>>>|
```

Result

The average marks of all subjects were calculated, and **Science** had the highest average of **87.5**.